

M Udhaya

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Professional Summary

Motivated Computer Science and Engineering (IoT) student with strong foundations in software development, IoT systems, and full-stack web technologies. Experienced in building real-time embedded systems, analytics dashboards, and AI-powered applications using React, Node.js, Flask, ESP32, and MongoDB. Skilled in problem-solving, collaborative development, and technical event leadership, with a CGPA of 9.563 and a strong interest in software engineering, machine learning, and scalable system design.

Education

Shiv Nadar University Chennai	2024 – 2028
B.Tech in Computer Science and Engineering (IoT)	CGPA: 9.563 (upto 3rd Semester)
Jeeva Velu International School, Tamil Nadu	2022 – 2024
CBSE Class XII: 93.6%	CBSE Class X: 98%

Technical Skills

Languages: C, C++, Python, Java, SQL, JavaScript

Web Development: React.js, Node.js, Express.js, Flask, REST APIs, Tailwind CSS

Databases: MySQL, MongoDB

IoT & Embedded Systems: ESP32, Arduino UNO, GPS, RTC Modules, Ultrasonic Sensors

Tools & Platforms: Git, GitHub, Streamlit, PCB Design, Arduino IDE

Core Concepts: DBMS, IoT Systems, Data Analytics, Machine Learning Basics

Projects

Wildlife Trade Risk Inspector *Flask, MongoDB, ML*

- Built an AI-powered ecological risk dashboard to analyze wildlife trade records and detect suspicious trade behavior patterns.
- Implemented Isolation Forest anomaly detection to generate poaching risk scores and identify high-risk exporting regions.
- Designed Species Comparison and Conservation Priority Index (CPI) modules for strategic conservation analysis.

Criclytics – IPL 2025 Analytics Dashboard *React, Node.js, MySQL*

- Developed a full-stack IPL analytics platform featuring player statistics, team analysis, and leaderboard dashboards.
- Built REST APIs with SQL JOIN queries and implemented dynamic routing using React Router.
- Created reusable UI components with responsive architecture and searchable dashboards.

GPS-Based Campus Shuttle Tracking System *ESP32, GPS, IoT*

- Designed a WiFi-based real-time shuttle tracking system using ESP32 and NEO-6M GPS modules.
- Developed an IoT workflow integrating GPS acquisition, WiFi communication, backend processing, and dashboard visualization.
- Implemented UART and HTTP-based communication for real-time location transfer.

Smart Water Tracker *Arduino UNO, RTC, Sensors*

- Built an offline hydration monitoring system with hourly reminders, inactivity alerts, and water-level tracking.
- Integrated RTC DS3231, HC-SR04 ultrasonic sensor, and buzzer-based notifications for event-driven monitoring.

Leadership & Activities

- Organized and coordinated technical and non-technical events including EduQuest 2025, Invente, and Vishranthi 2025.
- Served as EMCEE and event coordinator for multiple school and university events.
- Participated in Smart India Hackathon (SIH), SNUC Hacks, and university-level technical initiatives.
- Active NSS Volunteer with interests in technical problem-solving and collaborative development.

Achievements

- NPTEL AR/VR Certification – Elite Category (81%)
- School Topper in Core Subjects during CBSE academics
- Maintained a CGPA of 9.563 in B.Tech CSE (IoT)

Languages

English (Fluent), Tamil (Fluent), Hindi (Elementary)